

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

1. Consider the statement: $n\sqrt{n}$ is $O(n \log(n))$. True or False?

Justify your answer using the definition of $O(\cdot)$.

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2. Let $G = (V, E)$ be the graph with vertices $V = \{0, 1, 2, 3, 4, 5\}$ and edges $E = \{(0, 2), (0, 5), (1, 2), (1, 5), (0, 1), (2, 3), (0, 4), (1, 4), (3, 5), (2, 5)\}$.

- (a) Compute the breadth first search tree for G and for each vertex compute its degree. Show all steps.
- (b) What is the asymptotic cost of the vertex degree computation relative to the construction of the tree?

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3. Suppose we have a single resource and 8 requests to be scheduled, given by the list of tuples $L = [(0, 4), (9, 10), (7, 10), (3, 9), (8, 12), (5, 6), (7, 13), (7, 9)]$, where each tuple (s, f) gives the start time s and finish time f of each request.

- (a) Use the example to illustrate the scheduling of the requests such that no requests overlap and that the number of scheduled requests is maximal.
- (b) Explain why the greedy rule you applied leads to an optimal solution. Illustrate your explanations using the example.

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4. Let $G = (V, E)$ be a weighted graph with $V = \{0, 1, 2, 3, 4, 5\}$ and the edge between i and j with weight w represented by the triple (i, j, w) in $E = \{(0, 5, 5), (1, 3, 8), (1, 2, 6), (2, 5, 5), (0, 2, 1), (3, 5, 4), (4, 5, 7), (3, 4, 5), (1, 4, 5)\}$.

- (a) Construct the minimum spanning tree with Prim's algorithm. Show all steps.
- (b) When implementing Prim's algorithm, explain how the choice of the data structure leads to an $O(m \log(n))$ asymptotic cost for graphs of n vertices and m edges.

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5. Consider the sequence $S = 10, 56, 91, 42, 21, 68, 46, 67$.

- (a) Compute the number of inversions in this sequence via a divide-and-conquer algorithm. Show all steps.
- (b) In the worst case, a sequence of n elements may have as many as $O(n^2)$ inversions. Explain why divide-and-conquer leads to a subquadratic cost.

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6. Let C be the product of $A = \begin{bmatrix} 7 & 8 \\ 6 & 9 \end{bmatrix}$ with $B = \begin{bmatrix} 2 & 5 \\ 8 & 3 \end{bmatrix}$.

- (a) Use Strassen's method to compute C . Show all steps.
- (b) When using Strassen's method to multiply two n -by- n matrices, assume that the cost to add and subtract matrices is just n^2 and that the cost for the direct multiplication of two matrices is just n^3 . Use these assumptions to derive a threshold on n for which Strassen's method becomes more advantageous than the direct multiplication.

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